How accurate is altitude from pressure geolocators? A global assessment of three pressure-to-altitude conversions on the ground and in flight

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Abstract

Background: Pressure geolocators record the atmospheric pressure around all migratory birds throughout the year, and the altitude derived from it is used to study flight behaviour, habitat use and exposure to hazards. Converting pressure into altitude requires a reference level and an assumed temperature of the air column, yet the accuracy of the conversions used in practice has not been assessed systematically.

Methods: We compared three conversions: the standard atmosphere, widely used in geolocator studies; the same equation referenced to the ERA5 reanalysis, as in GeoPressureR up to v3.6; and a version adding the virtual temperature and the climatological lapse rate of the GPT3 troposphere model, the GeoPressureR default from v3.7.0. Assuming a perfect pressure reading, we evaluated them on the ground at 2,502 weather stations worldwide and 185 stations with surveyed barometer heights, and in flight at 648 radiosonde stations up to 6 km above the ground, weighting heights by the flight time of 274 tracked birds.

Results: On the ground, the standard atmosphere was off by tens of metres (median absolute station bias 23.9 m, temporal scatter 63.2 m), because weather moves the surface pressure. With the ERA5 reference, the median station had a constant bias of 2.5 m and a scatter of 3.6 m, rising to tens of metres in mountains.

In flight, both earlier conversions placed birds too low by about 1% of their height above the ground, because they assume a column that is too cold; over the heights birds fly, the mean absolute error was 41.2 m with the standard atmosphere and 16.4 m with ERA5. The temperature-corrected conversion removed most of this bias (-3.2 m instead of -11.8 m) and reduced the mean absolute error to 13.1 m, with little change on the ground; its remaining scatter follows the day-to-day temperature profile.

Conclusions: A reanalysis reference makes pressure-derived ground altitude accurate and precise to a few metres, while the assumed column temperature sets the accuracy of flight heights. Our error estimates for each conversion allow altitudes from past and future geolocator studies to be interpreted with their uncertainty.

Keywords: biologging, barometric altimetry, flight altitude, ERA5, radiosonde, multi-sensor geolocator, GeoPressureR

1 Background

The altitude of a bird matters both on the ground and in flight. On the ground, altitude constrains where the bird is, refining a position that pressure alone resolves only coarsely [1], and it describes the habitat used, as in the uphill roosting movements of Alpine-breeding wheatears [2]. In flight, altitude reveals how birds use winds and temperatures aloft and cross ecological barriers [3, 4], and how often they fly through the height band of wind turbines [5–7].

Small migrants are too light to carry GPS, but multi-sensor geolocators weighing 1–1.5 g [8, 9] now record atmospheric pressure along entire migrations [10] and throughout the year [9, 11]. They have revealed flights at several thousand metres and large altitude changes between night and day [3, 12, 13]. On larger birds that carry GPS, barometric heights are less noisy than GPS heights from standard fixes, but they carry a systematic bias [14, 15]. Pressure, however, is not altitude. Converting one into the other needs a reference level, where both pressure and altitude are known, and a temperature for the air between that reference and the bird. Both vary in space and time, and errors in either propagate into flight-height distributions and the ecological conclusions drawn from them [16, 17].

Geolocator studies have converted pressure in three ways. Many have used the barometric equation of the standard atmosphere, as in aircraft altimeters, with a fixed reference at sea level and a fixed temperature profile [18], and report altitudes above sea level [9, 12, 13, 19, 20]; the pamlr toolbox implements this equation [21]. These authors noted that the weather can shift such altitudes by several tens of metres [9], and by hundreds at high altitude over warm air [12]. A second approach replaces the standard sea-level pressure by that of a reanalysis at the bird's approximate position [22], or, for seabirds, by the pressure recorded when the bird rests on the sea surface

[5, 6], but such calibrations are not available for most terrestrial migrants. The third takes the reference at the ground from a weather model, which gives the pressure and temperature at the surface anywhere and at any time. GeoPressureR, which reconstructs the trajectories of birds carrying pressure geolocators [1, 23], uses the surface pressure, 2 m temperature and orography of the ERA5 reanalysis [24] at the bird's position and hour [2], and Schaub et al. [15] used ERA5 with raptor GPS-barometer tags. Whatever the reference, most of these conversions keep a fixed lapse rate above it, usually that of the standard atmosphere.

The accuracy of these conversions is poorly known. Field comparisons are local and short, and mix the errors of the sensor, the calibration, the terrain model and the conversion [6, 14, 15]. Validations of reanalysis surface pressure, conversely, report errors of about 1 hPa, i.e. 8 m [25, 26], but at the station height only: they do not test the temperature assumed above the ground, and they depend on barometer heights that are wrong at many synoptic stations [27, 28]. Other fields have faced the same problem. GNSS meteorology brings model pressure to the height of each station with a virtual temperature and a lapse rate that vary with place and season [29, 30], and aviation corrects altimeter readings when the air is colder than the standard atmosphere [31] or derives geometric altitude from barometers and ERA5 temperature profiles [32].

Here, we evaluate three conversions as used with geolocators, the standard atmosphere, the ERA5 reference with the standard lapse rate, and the same reference with the virtual temperature and climatological lapse rate of GNSS meteorology, against references of known height that span the terrain, climates and heights birds experience: weather stations worldwide, stations with surveyed barometer heights, and radiosondes up to 6 km above the ground. We ask how accurate (constant bias) and precise (scatter over time) each conversion is, on the ground and in flight, and what explains their errors. The third conversion borrows an established correction rather than proposing a new method; we test whether it is worth adopting, since it costs no more than the second and became the GeoPressureR default.

2 Methods

2.1 Three conversions from pressure to altitude

Pressure decreases with height because each layer of air carries the weight of the air above it. Combining this hydrostatic balance with the ideal gas law gives the hypsometric equation, which holds for any temperature profile [33]:

$$z = z_0 + \frac{R\,\bar{T}_v}{g\,M}\,\ln\frac{p_0}{p}, \tag{1}$$

with z the altitude (m) at pressure p, z_0 and p_0 the altitude and pressure of a reference level, $g = 9.807$ m s^{-2}, $R = 8.314$ J mol^{-1} K^{-1}, $M = 0.02896$ kg mol^{-1}, and $\bar{T}_v$ the

mean virtual temperature of the air column between the reference level and the bird. The virtual temperature $T_v = T(1 + 0.608\,q)$, with q the specific humidity, accounts for moist air being lighter than dry air. Because the height above the reference level is proportional to $\bar{T}_v$, an error in the assumed mean temperature becomes the same relative error in height: a column assumed 3 K (1%) too cold places a bird flying 2 km above the reference level about 20 m too low. The column temperature is never known exactly, so every conversion approximates it, most often by a temperature T_0 at the reference level that changes linearly with height at a lapse rate L, which gives the barometric equation:

$$z = z_0 + \frac{T_0}{L}\left[\left(\frac{p}{p_0}\right)^{-RL/(gM)} - 1\right]. \tag{2}$$

The three conversions differ in their reference level and in the column temperature they assume (Figure 1):

1. **Standard atmosphere.** The reference is sea level ($z_0 = 0$, $p_0 = 1013.25$ hPa) with $T_0 = 15$ °C and $L = -6.5$ K km^{-1}, in dry air [18]. It needs no external data, but ignores the weather: a change of 10 hPa in surface pressure shifts the altitude by about 80 m, and a column 20 K warmer or colder than assumed changes heights by about 7%.
2. **ERA5, standard lapse rate.** p_0, T_0 and z_0 are the surface pressure, 2 m temperature and orography of the ERA5 grid cell (0.25°) and hour of each pressure measurement, with $L = -6.5$ K km^{-1} in dry air. This was the GeoPressureR formula up to v3.6 (`altitude_formula = "standard"`).
3. **ERA5, virtual temperature and GPT3 lapse rate.** The same reference level, with T_0 the ERA5 2 m virtual temperature, computed from the 2 m dewpoint and surface pressure [34], and L the temperature lapse rate of GPT3, the empirical troposphere model of GNSS meteorology [30]. GPT3 gives the lapse rate on a 1° grid as a mean with annual and semi-annual cycles, derived from reanalysis profiles; we interpolate it bilinearly at the position and day of year. It is shallow or positive where surface inversions are frequent, e.g. in continental and polar winters. This is the GeoPressureR default from v3.7.0 (`altitude_formula = "virtual"`).

In tables, we abbreviate them as Standard atm., ERA5 standard and ERA5 virtual + GPT3. None of the three is fitted to the references used here. The ERA5 variables are read as GeoPressureR does, from the ECMWF ARCO archive, at the nearest grid cell and hour. The third conversion needs one more ERA5 variable (2 m dewpoint) and a 1° lookup table, so it is no more expensive to compute along a trajectory than the second. Pressure-level data, which would give the actual temperature profile, are not in the ERA5 single-levels archive and would require far more data per point.

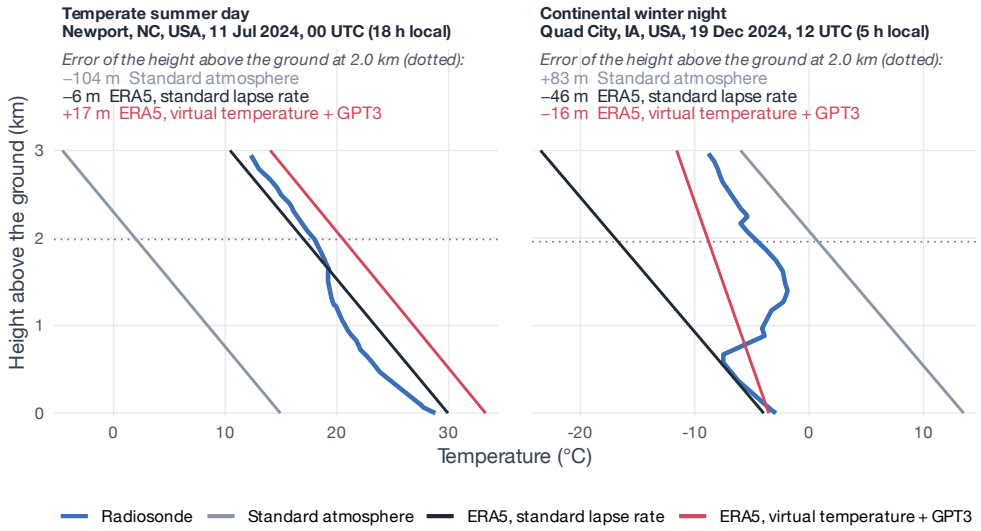


Figure 1: Column temperature assumed by the three conversions against the temperature measured by a radiosonde, for two typical soundings. The height above the ground is proportional to the mean temperature of the column below the bird (Eq. 1): a column assumed too cold places the bird too low, one assumed too warm, too high. The standard atmosphere starts from 15 °C at sea level whatever the weather; the two ERA5 conversions start from the ERA5 2 m temperature, or 2 m virtual temperature (up to about 3 K warmer in moist air), and decrease at the standard or the GPT3 lapse rate. Text: error of the height above the ground at the level closest to 2 km (dotted). Each sounding is, within its class, the one whose three errors at that level are jointly closest to the class medians: (left) Köppen C, June–August, 07–19 h local solar time (11,471 soundings; medians −104, −5 and +17 m); (right) Köppen D, December–February, at night (5,812 soundings; medians +84, −48 and −18 m); southern-hemisphere seasons shifted by six months.

2.2 Reference data

An ideal reference would record pressure continuously at a height known to within a metre, cover the terrain and climates birds experience, and be independent of ERA5. No dataset meets all criteria, so we use four, each for the question it answers best (Table 1, Figure 2; Additional file 1: S1):

- **HadISD** [35, 36], a quality-controlled subset of the Integrated Surface Database [37]: station pressure of synoptic weather stations worldwide, 2022–2024. It is the only global ground reference, but ERA5 assimilates these pressures, and the listed elevation is not always the barometer height. We thinned it to one station per 2°×2° cell, plus all stations above 1000 m, weighted each cell equally, and screened stations

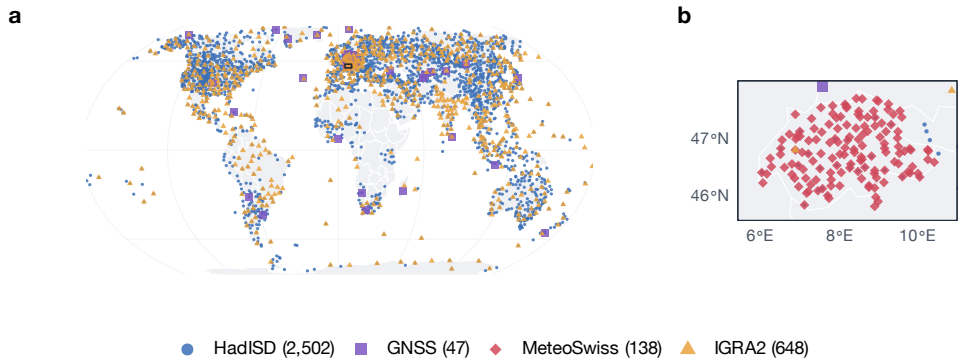


Figure 2: Reference stations used after quality control: (a) worldwide and (b) in Switzerland (box in a), where the MeteoSwiss stations are. Numbers of stations in brackets.

Table 1: Reference data, 2022–2024 (radiosondes 2023–2024): stations used after quality control, source of the barometer height, and whether ERA5 assimilates the pressure.

Reference	Type	Stations	Barometer height	In ERA5	Use
HadISD	synoptic stations	2,502	listed, screened	yes	ground, global
GNSS	geodetic stations	47	surveyed	no	ground, surveyed
MeteoSwiss	Swiss automatic stations	138	published	mostly	ground, mountains
IGRA2	radiosondes	648	not needed	yes	flight

for wrong barometer heights (step changes, unreliable barometers and implausible constant offsets; Additional file 1: S1), keeping 2,502 stations.

- **GNSS stations** of the IGS and EUREF networks with a barometer (47 stations): their barometer heights are surveyed relative to an antenna known to the millimetre, and their pressure is not assimilated by ERA5.
- **MeteoSwiss** automatic stations (138 stations), which publish the height of each barometer, in the steep Alpine terrain where ground errors are largest.
- **IGRA2 radiosondes** [38, 39], 648 stations, 2023–2024: pressure and height at each level up to 6 km above the ground, the only reference in flight.

2.3 Error metrics

We gave each conversion the pressure measured by the reference barometers, as if a geolocator had recorded it, and compared the altitude returned with the known height. We thus evaluate the conversion only, assuming a perfect pressure sensor.

Table 2: Ground error of the three conversions (m), over the stations of each reference: median and 90th percentile (p90) of the station |bias| (accuracy) and of the station SD (precision), and mean absolute error (MAE). HadISD statistics weight each 2°×2° cell equally.

Conversion	Reference	Stations	|bias| median	|bias| p90	SD median	SD p90	MAE
Standard atm.	HadISD	2,502	23.9	50.4	63.2	96.6	59.3
Standard atm.	GNSS	47	21.8	54.1	74.0	95.0	61.8
Standard atm.	MeteoSwiss	138	41.1	54.9	68.0	78.5	68.8
ERA5 standard	HadISD	2,502	2.5	11.4	3.6	7.0	6.1
ERA5 standard	GNSS	47	2.3	8.7	3.0	7.0	5.4
ERA5 standard	MeteoSwiss	138	2.9	9.7	5.4	13.5	6.9
ERA5 virtual + GPT3	HadISD	2,502	2.6	11.0	3.7	7.2	6.1
ERA5 virtual + GPT3	GNSS	47	2.4	7.3	2.9	7.5	5.3
ERA5 virtual + GPT3	MeteoSwiss	138	2.6	7.6	5.8	15.0	7.0

On the ground, the error is the retrieved altitude minus the barometer height. At each station, it is split into its mean, the **bias**, which sets the **accuracy** of absolute altitude, and the standard deviation (SD) around that mean, which sets the **precision** of altitude changes at one site. Across stations, we report the median and 90th percentile (p90) of the |bias| and of the SD, and the mean absolute error (MAE), which combines both.

In flight, the error at each radiosonde level is the error minus that at the surface of the same sounding: it measures the error of the **height above the ground**, the quantity left when a flight is referenced to the bird's own ground before departure and after arrival, and it cancels any error in the station elevation. Bias and SD are computed per station and height bin (0.1–0.25, 0.25–0.5, 0.5–1, 1–1.5, 1.5–2 km, then 1 km bins to 6 km). To summarise over the heights birds actually fly, we weighted each bin by the share of flight time of 274 tracked birds from the GeoLocator master data package, whose flight time is half below 560 m and 90% below 2,320 m above the ground (Figure 4 c). Radiosonde statistics weight each 2°×2° cell equally.

3 Results

3.1 Accuracy and precision on the ground

The ERA5 reference reduced the ground error tenfold (Table 2, Figure 3). With the standard atmosphere, the median station was biased by 23.9 m and scattered by 63.2 m over time, the signature of a reference that ignores the weather: the bias follows the mean departure of a site's surface pressure and temperature from the standard atmosphere, and the scatter follows passing weather systems. With ERA5, the median station had a constant bias of 2.5 m and a scatter of 3.6 m. The bias was skewed: at 10% of the stations it exceeded 11.4 m, mostly where the station lies far above or below

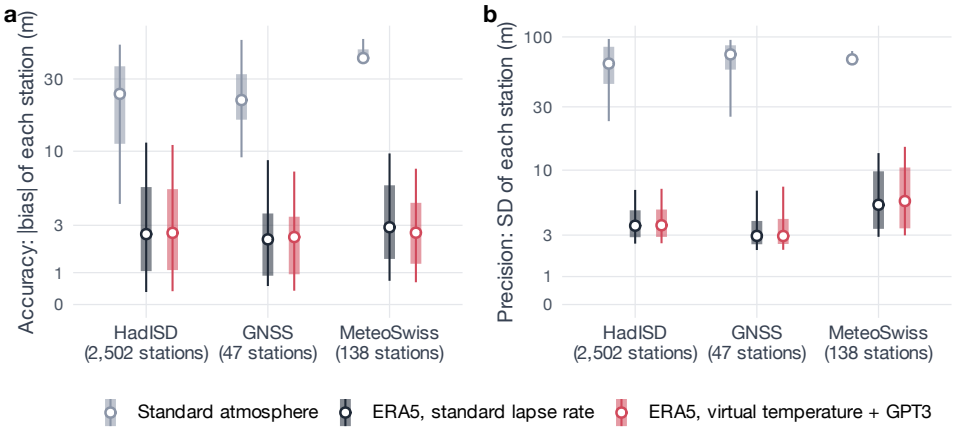


Figure 3: Ground error of the three conversions at the three ground references, for a perfect pressure reading: (a) absolute value of the mean error of each station (|bias|, accuracy of absolute altitude) and (b) standard deviation of the error of each station over time (SD, precision of altitude changes at one site), on a log scale. Circle: median; thick bar: 25–75%; thin bar: 10–90% of stations (HadISD: each 2°×2° cell weighted equally).

the 0.25° ERA5 orography, through which the surface pressure must be extrapolated. The median |bias| rose from 1.9 m in flat terrain (sub-grid standard deviation of the ERA5 orography below 20 m) to 5.9 m in mountains (above 300 m), and the median SD from 3.1 to 8.2 m. Season, time of day and climate mattered little (Additional file 1: S2).

The temperature correction hardly changed the ground altitude, because it only acts over the elevation gap between the station and the ERA5 orography: the median |bias| was 2.6 m and the MAE 6.1 m, against 6.1 m. The three references agreed (Table 2): the GNSS and MeteoSwiss stations, with surveyed or published barometer heights, gave the same accuracy and precision as HadISD.

3.2 Accuracy and precision in flight

In flight, the error grew with height for all three conversions (Figure 4, Table 3). Both earlier conversions placed birds too low, by about 1–1.5% of the height above the ground: the ERA5 conversion by 14.7 m at 1–1.5 km and 35.1 m at 2–3 km. The standard atmosphere had a similar mean bias but a much larger scatter, because its column temperature ignores the actual surface temperature, which departs by tens of kelvins from 15 °C over the seasons and latitudes; over the heights birds fly, its MAE was 41.2 m, against 16.4 m with ERA5. For the ERA5 conversion, the error expected from the difference between the assumed and the observed mean column temperature explained 94% of the variance of the in-flight error (Additional file 1: S2): the in-flight

Table 3: Error of the height above the ground in flight (m), as bias / RMSE in four height bins and as bias / MAE over the heights birds fly (bird-weighted), pooled over all radiosonde levels with each 2°×2° cell weighted equally.

Conversion	0.1–0.25 km	0.5–1 km	1–1.5 km	2–3 km	Bird-weighted
Standard atm.	-2.4 / 14.6	-9.6 / 40.1	-8.3 / 69.7	-5.5 / 128.2	-12.2 / 41.2
ERA5 standard	-3.1 / 12.0	-8.0 / 17.0	-14.7 / 25.5	-35.1 / 59.0	-11.8 / 16.4
ERA5 virtual + GPT3	-2.4 / 11.9	-2.9 / 15.0	-3.8 / 19.6	-4.7 / 42.9	-3.2 / 13.1

error is an error of the assumed temperature profile, not of the reference level. The bias was largest in continental and polar winters, when strong surface inversions sit under much warmer air (Figure 5), and at night (Additional file 1: S2).

The virtual temperature and GPT3 lapse rate removed most of this bias, from -14.7 to -3.8 m at 1–1.5 km and from -35.1 to -4.7 m at 2–3 km, and halved the bias in continental and polar winters (Figure 5). Over the heights birds fly, the bias dropped from -11.8 m to -3.2 m and the MAE from 16.4 m to 13.1 m. The precision, however, hardly changed: the SD was 19.3 m at 1–1.5 km, against 20.8 m with the standard lapse rate. Part of the remaining error changes sign between day and night: at 2–3 km in the summer half, the bias was +16.0 m by day and -18.4 m at night (Figure 1). No lapse rate set from the date and place can follow the day-to-day temperature profile. Lapse rates fitted to the radiosondes themselves, constant or varying with season and latitude, did no better on held-out stations (MAE 13.0 m; Additional file 1: S5).

3.3 Reliability of the references

Three checks support that these figures describe ERA5, not the references (Additional file 1: S4). First, HadISD's errors partly come from its listed elevations: at 32 Swiss stations present in both datasets, the HadISD bias followed the difference between the listed elevation and the published barometer height almost one for one ($r = 0.93$), and the median |bias| dropped from 7.2 to 3.8 m with the barometer heights. The HadISD figures are therefore, if anything, pessimistic. Second, GNSS stations, never assimilated by ERA5, were fitted as well as assimilated HadISD stations in the same terrain, and isolated HadISD stations, whose observations weigh most in the analysis, did not fit better than stations in dense networks. Third, the screen of barometer heights only affected the tail of the HadISD errors, not the median.

4 Discussion

4.1 What accuracy and precision to expect

Table 4 summarises the error a user can expect from each conversion. Two components of the conversion behave differently. The **reference level** sets the error on the ground:

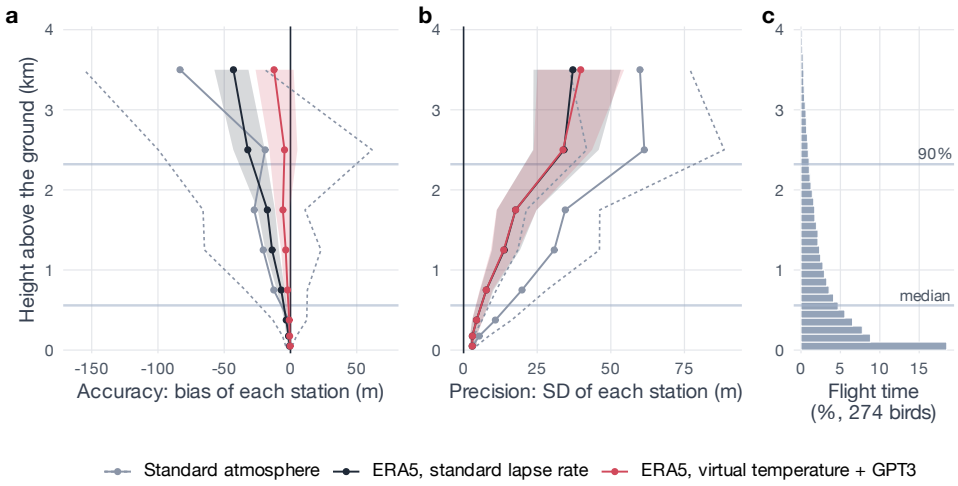


Figure 4: Error of the height above the ground in flight against height above the ground, for the three conversions: (a) mean error of each radiosonde station (bias, accuracy; negative: bird placed too low) and (b) its standard deviation over soundings (SD, precision), per height bin. Line and points: median over stations; shaded band (dashed outline for the standard atmosphere): 25–75% of stations; each 2°×2° cell weighted equally. (c) Share of the flight time of 274 tracked birds by height above the ground; the horizontal lines in all panels mark the median (560 m) and 90th percentile (2,320 m) of flight height.

replacing the standard atmosphere with ERA5 reduced it from tens of metres to a few, and what remains is mostly a constant offset per site from the terrain that the 0.25° grid cannot resolve, while hour, season and weather hardly matter. Altitude changes at one site are therefore more reliable than absolute altitudes, and comparing absolute altitudes between distant sites, or with a fine-scale digital elevation model, should allow for tens of metres in mountains. The **column temperature** sets the error in flight. Because the earlier conversions assume a column that is usually too cold, flight heights derived with them are conservative, about 1–1.5% too low, twice as much in continental and polar winters and at night, when surface inversions are frequent. The standard atmosphere adds a large seasonal and geographic scatter on top. The temperature-corrected conversion removes most of the bias, but the scatter that follows the day-to-day temperature profile remains, about 20 m at 1–1.5 km and 43 m at 2–3 km.

The pressure sensor adds its own error, which we did not evaluate. A pressure error δp shifts the altitude by $R T_v \, \delta p / (g \, M \, p)$, about 8.4 m hPa^{-1} near sea level and 10 m hPa^{-1} at 2 km. The barometers of geolocators typically specify an absolute accuracy of the order of 1 hPa and a much finer relative accuracy. An absolute offset therefore matters most for ground altitude, where it can exceed the conversion error in flat

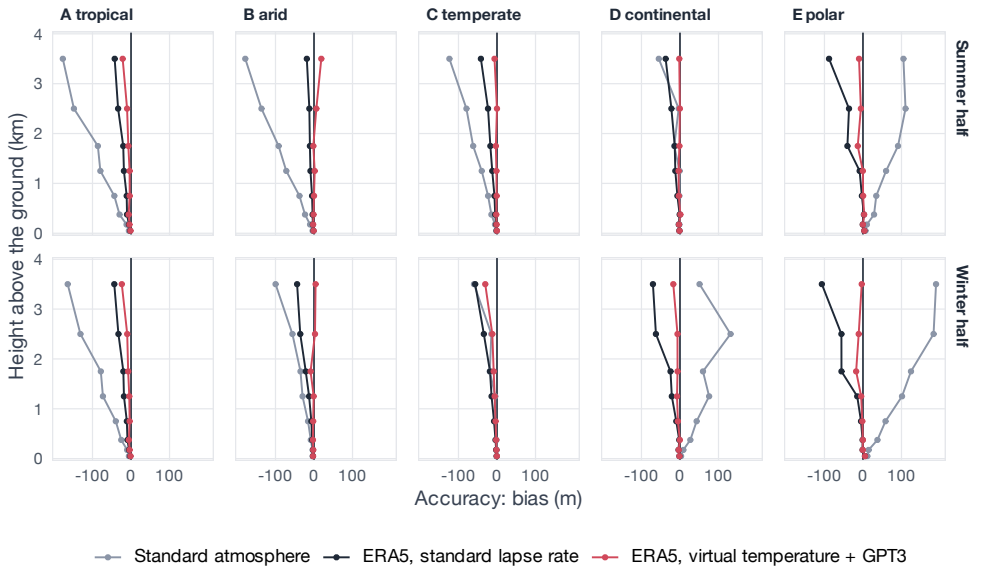


Figure 5: Bias of the height above the ground in flight against height above the ground, for the three conversions, by main Köppen-Geiger climate zone (columns) and half-year (rows; summer half: 21 March–22 September in the northern hemisphere, shifted by six months in the southern). Pooled over the radiosonde levels of each class, each 2°×2° cell weighted equally; classes with fewer than three stations omitted.

terrain; it cancels from flight heights referenced to the bird's own ground and from altitude changes, which depend only on the sensor's drift and relative error. GeoPressureR positions are not affected by a constant offset, because they are computed from pressure changes [1].

4.2 Comparison with other studies

Our pooled ground RMSE with ERA5 (9.6 m, about 1.2 hPa) is close to the RMSE of about 1 hPa reported for ERA5 surface pressure at weather stations in China [26] and for ERA-Interim at GNSS stations [25], a study which, as in our screen, had to discard stations whose pressure records showed offsets. On grounded raptors, Schaub et al. [15] found ERA5-referenced barometric heights 5–15 m below the terrain (median per tag model), more than our ground offsets. Their figures also include the tag's own offset and the error of the terrain model, which our references exclude, and the bias appeared smaller in flight, which points to the tag on a resting bird rather than to the conversion. Lato et al. [14] found barometric heights on average 1–2 m below those of a laser altimeter at 2–50 m above take-off; at such heights, the conversion error is small (bias -1.4 m below 100 m with ERA5), so their figure mostly reflects the sensor.

Table 4: Expected error (m) of pressure-derived altitude for a perfect pressure reading, and what the pressure sensor adds (an absolute offset shifts altitude by about 8 m hPa^{-1}). Ground: HadISD reference set; flat terrain and mountains: sub-grid standard deviation of the ERA5 orography below 20 m and above 300 m; change at one site: median station SD. Flight: error of the height above the ground.

Quantity	Standard atm.	ERA5 standard	ERA5 virtual + GPT3	Sensor adds
Ground, flat terrain (MAE)	58.2	4.6	4.7	offset
Ground, mountains (MAE)	65.4	13.4	13.4	offset
Change at one site (SD)	63.2	3.6	3.7	relative error
Flight, 1–1.5 km (RMSE)	69.7	25.5	19.6	drift, relative error
Flight, 2–3 km (RMSE)	128.2	59.0	42.9	drift, relative error
Flight, bird-weighted (MAE)	41.2	16.4	13.1	drift, relative error

Comparisons between barometric and GPS heights in flight [6, 15] cannot identify which estimate is right, so they cannot test the height-dependent bias we find.

The closest earlier assessment of the in-flight error is that of Lindström et al. [12], who compared standard-atmosphere altitudes of great snipes with reanalysis geopotential heights along approximate routes: the standard atmosphere placed the birds about 80 m too low at 3 km over Europe in autumn and about 400 m too low at 7 km over Africa in spring, in the warm air of the subtropical highs. Our results generalise this observation: the error of the standard atmosphere follows the temperature of the column, so that it is small on average over seasons and climates but scatters widely (SD 128 m at 2–3 km). None of the corrections we tested is new. Reducing pressure with a virtual temperature and a climatological lapse rate is routine in GNSS meteorology [29, 30], and aviation corrects altimeters for cold air [31]; what we add is an assessment of how much such a correction changes the altitude of birds, on the ground and over the heights they fly. Barometric altitude computed with the full ERA5 temperature profile on pressure levels agreed with post-processed GNSS altitude to a few metres (SD of 4–6 m over nine test flights of an aircraft) [32], which shows what knowing the actual profile can achieve, at the cost of far more data per point.

4.3 Recommendations

For ground altitude and flight heights, a reanalysis reference should be preferred to the standard atmosphere, which is off by tens of metres on the ground and does not account for the temperature of the column in flight. Flight heights computed with the standard atmosphere or with the ERA5 standard lapse rate can be read as about 1% too low on average, with a larger error in winter and at night; with the standard atmosphere, a single flight can moreover be off by several percent either way, depending on the temperature of the air it crossed. Single-levels ERA5 should be used: ERA5-Land, despite its finer 0.1° grid, was about four times less accurate on the ground, because its surface pressure is not consistent with the orography it publishes, which

shifts every altitude by a fixed offset that depends on the place (Additional file 1: S3). Older tracks are somewhat less accurate: for the same stations, the median ground |bias| was 5.9 m in 1990 against 4.1 m in 2024, likely because fewer observations constrained ERA5.

4.4 Limitations

These are errors of the conversion, for a perfect pressure reading at a known position. The radiosonde heights are themselves mostly computed from the sonde's pressure, temperature and humidity, so the in-flight error measures the error of the column temperature assumed by each conversion relative to the sonde's; the ERA5 column at the launch site is used and balloon drift is ignored. The stations with documented heights are few and unevenly spread, and all references are on land. Where stations are sparse (parts of Africa, South America and Siberia), ERA5 is less constrained and the error is probably larger than our global figures. Finally, the bird-weighted metric uses a flight-height distribution derived with an ERA5 conversion; its errors of about 1% do not affect the weights.

5 Conclusions

The error of pressure-derived altitude has two parts with different causes. The reference level sets the error on the ground: the median station was off by 23.9 m with the standard atmosphere, which ignores the weather, and by 2.5 m with the ERA5 reference, a constant offset per site that grows to tens of metres in mountains. The assumed column temperature sets the error in flight. Over the heights birds fly, the mean absolute error of the height above the ground was 41.2 m with the standard atmosphere and 16.4 m with ERA5 and the standard lapse rate, both placing birds about 1% too low on average. The virtual temperature and GPT3 lapse rate reduced it to 13.1 m and the bias from -11.8 to -3.2 m, at no extra cost and without changing ground altitudes; it is now the default in GeoPressureR. The remaining in-flight scatter, an RMSE of about 20 m at 1–1.5 km and 43 m at 2–3 km, follows the day-to-day temperature profile and is the limit of any conversion based on surface data alone.

6 Declarations

Ethics approval and consent to participate. Not applicable: no animals were handled for this study; the bird flight heights come from published tracking data.

Consent for publication. Not applicable.

Availability of data and materials. Code, derived tables, figures and the downloaded data needed to rerun the analysis (all but HadISD) are archived on Zenodo [40] and at https://github.com/GeoPressure/altitude-validation, with an online report of all results. The conversions are implemented in GeoPressureR [41]. HadISD,

IGRA2, GNSS, MeteoSwiss and ERA5 data are openly available from their providers (Additional file 1: S1).

Competing interests. The authors declare that they have no competing interests.

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Authors' contributions. RN designed the study, performed the analysis and wrote the manuscript. MG contributed to the implementation of the conversions in GeoPressureR and GeoPressureAPI and revised the manuscript. [To be confirmed.]

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